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Network Analysis

Second Assignment

1 Overview

The values reported in the following part represent the **default parameters for our simulation**, chosen based on several experimental configurations aimed at achieving **effective information propagation across the network**.
The underlying idea was that for a node to adopt the information, the following condition must be satisfied:

$$\frac{\text{honest neighbors} + \text{malicious neighbors}}{\text{total neighbors}} > \theta$$

Two metrics proved particularly important when tuning these values:

- **Average degree:** Knowing that the average node has approximately **14 neighbors** helps to estimate the expected denominator in the propagation formula.
- **Number of nodes:** This provides insight into how many **gossip starters and malicious nodes** are needed to reach a sufficient fraction of the network.

Therefore, it is essential to **identify suitable values for** θ, as well as for the **number of gossip starters and malicious nodes**, to ensure that propagation can actually occur.
If the number of gossip starters and malicious nodes is **too low relative to the average degree and the network size**, especially in large networks, the threshold condition may never be exceeded, thus **hindering the spread of information**.

Based on the observations above, the **optimal parameters selected** for the network are as follows:

- θ: **0.15** (Fixed for all the nodes)
- Gossip starters: **30**
- Malicious starters: **50**
- Number of nodes: **5,908**
- Average degree: **14**

2 Simulations

In this section, we explore **different simulation configurations** where **two parameters are held constant**, and the **third is varied** across a predefined range. This setup enables us to **isolate and analyze the individual effect** of each parameter on the diffusion process. All results displayed in the plots refer to the **final state of the simulation** for each value of the varying parameter.

Example Varying the Number of Gossiper:

- **Fixed parameters:** $\theta = 0.15$, Malicious $= 50$
- **Gossipers tested:** 5, 10, 50, 100, 250, 500, 1000, 3000, 5000, 5800

The plots display three lines, each corresponding to a different fixed parameter configuration: one for **theta**, one for **malicious nodes**, and one for **gossipers**. Along each line, **dots are placed at specific intervals** these represent the final number of nodes in a certain state for each configuration, where the **number of gossipers is varied**.
Each set of dots aligned vertically (i.e., sharing the same x-axis value) corresponds to the final state of the same simulation, under one specific configuration. The **sum of the three vertical dots (one per line)** always equals the **total number of nodes in the network**, which is **5,908**. This allows for an intuitive visual comparison of how the different states are distributed at the end of each simulation.
This type of analysis is particularly useful as it **highlights how the network's behavior evolves** in response to an increase or decrease in a single parameter at a time. It helps us gain **deeper insights into the sensitivity, robustness, and limitations of the system** under different conditions.

Default simulation values:

- θ: **0.15**
- Gossip starters: **30**
- Malicious starters: **50**

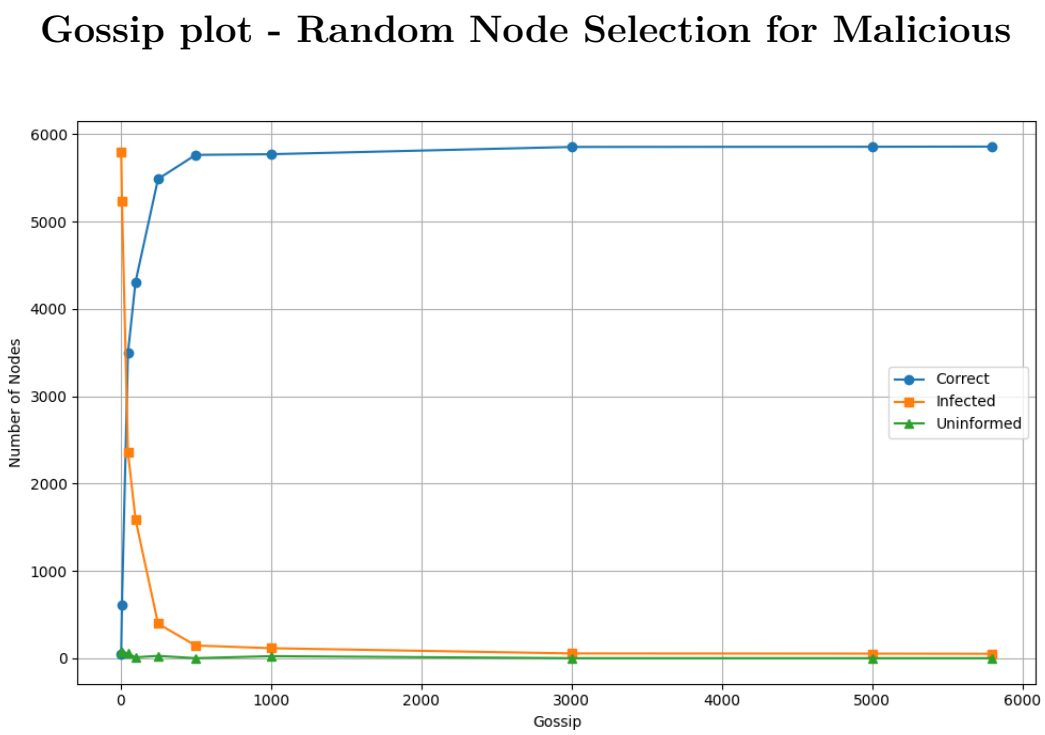


Figure 1: Information propagation with fixed θ and malicious starters, varying the number of gossip starters.

From the graph, we can observe that reaching approximately **half of the network with the correct message requires only 50 gossipers**, which **matches the number of malicious nodes** in the current configuration. However, achieving a **significantly higher percentage of correct diffusion** specifically **5487 out of 5908 nodes** requires just **250 gossipers**. This is a **surprisingly low number** considering the size of the network, and it is only **five times greater than the number of malicious nodes**.

Default simulation values:

- θ: **0.15**
- Gossip starters: **30**
- Malicious starters: **50**

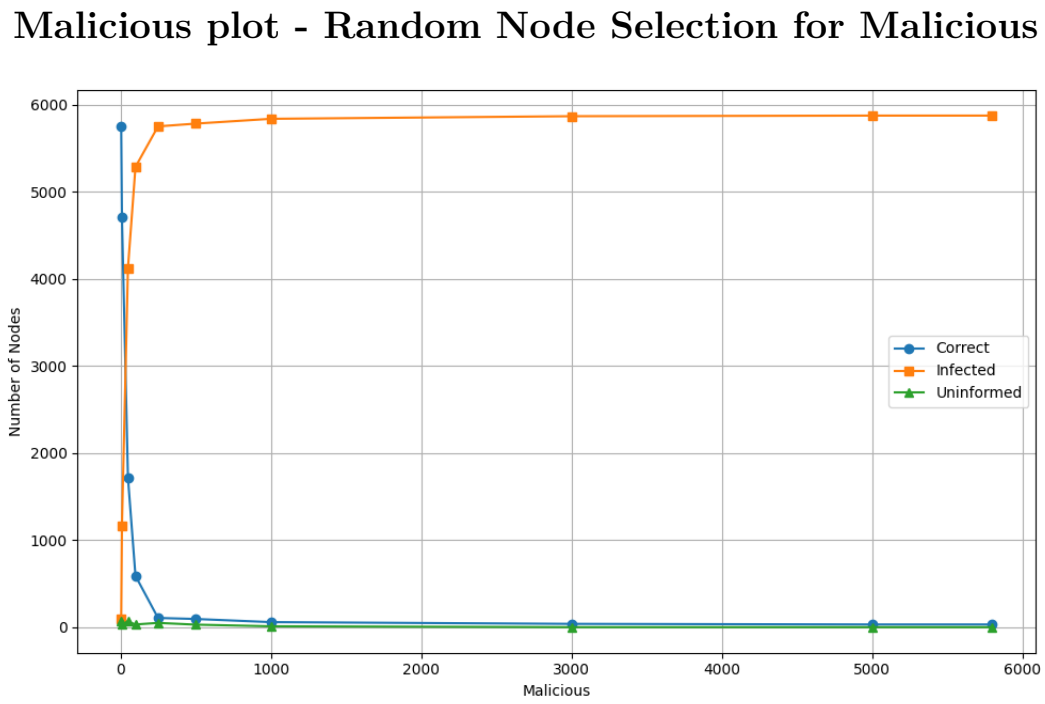


Figure 2: Information propagation with fixed θ and gossip starters, varying the number of malicious starters.

In the malicious plot instead we can observe that even in this case it takes just **50 malicious nodes** to reach the majority of the network, **similarly to the gossip configuration**. However, a **key difference** lies in the **number of nodes influenced** by those 50 malicious individuals: there is an **increase of approximately 600 nodes** carrying malicious information compared to the gossip scenario.
This behavior is mainly due to the **low default number of gossipers**, which makes the network more vulnerable to malicious diffusion. Therefore, we can conclude that **even fewer than 50 malicious nodes**, possibly **between 31 and 49**, **could be enough to reach the majority** in this configuration.
In order to achieve results comparable to those obtained in the gossip configuration, we need only **100 malicious nodes**. In fact, with this number, we reach **5288 out of 5908 nodes**, which indicates a very effective spread of tampered information.
As previously discussed, this outcome is expected given the **low number of gossip starters**, which makes the network more susceptible to malicious influence.

Default simulation values:

- θ: **0.15**
- Gossip starters: **30**
- Malicious starters: **50**

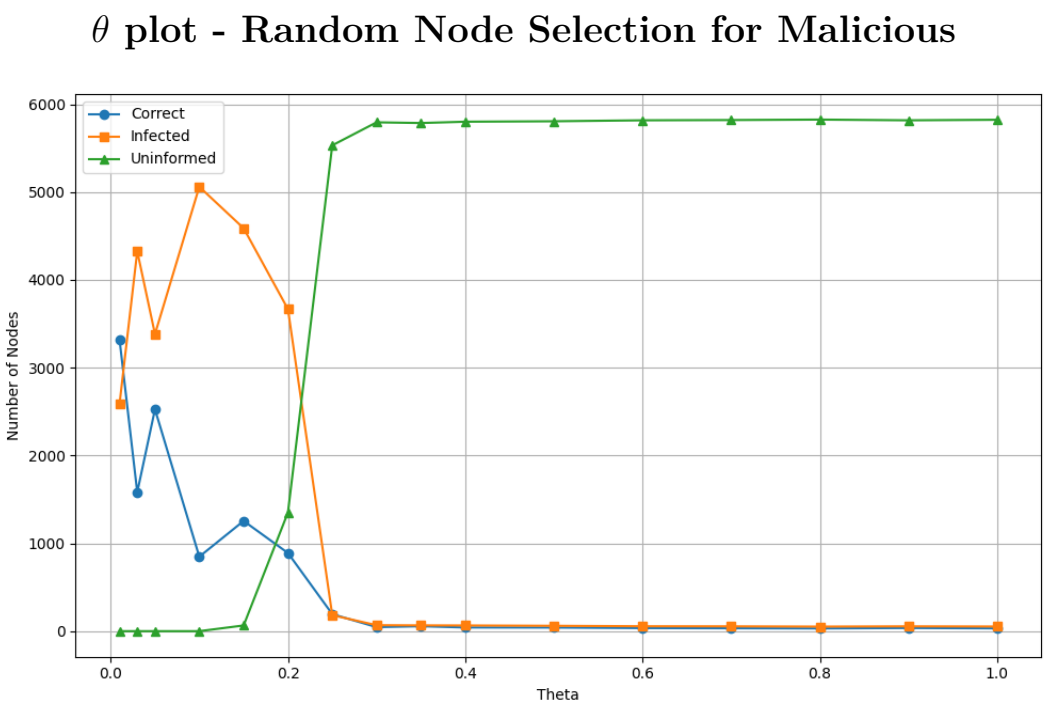


Figure 3: Information propagation with fixed malicious starters and gossip starters, varying θ.

For the θ plot, we observe that when θ is equal to **0.25 or higher**, the message **barely propagates** through the network. This behavior is expected due to the relatively **small number of active nodes** (gossip + malicious) and the **average degree of nodes** in the graph.
Conversely, there is a **significant improvement in propagation** when θ is equal to or below **0.20**. In particular, for values equal to or below **0.10**, we consistently achieve **full propagation** across the entire network.
However, to simulate a more **realistic diffusion process**, a value of $\theta = \mathbf{0.15}$ proves to be a **good compromise**: the message **spreads widely**, but **not to every node**, leaving a few **uninformed outsiders**, which better reflects **real-world scenarios**.

Default simulation values:

- θ: **0.15**
- Gossip starters: **30**
- Malicious starters: **50**

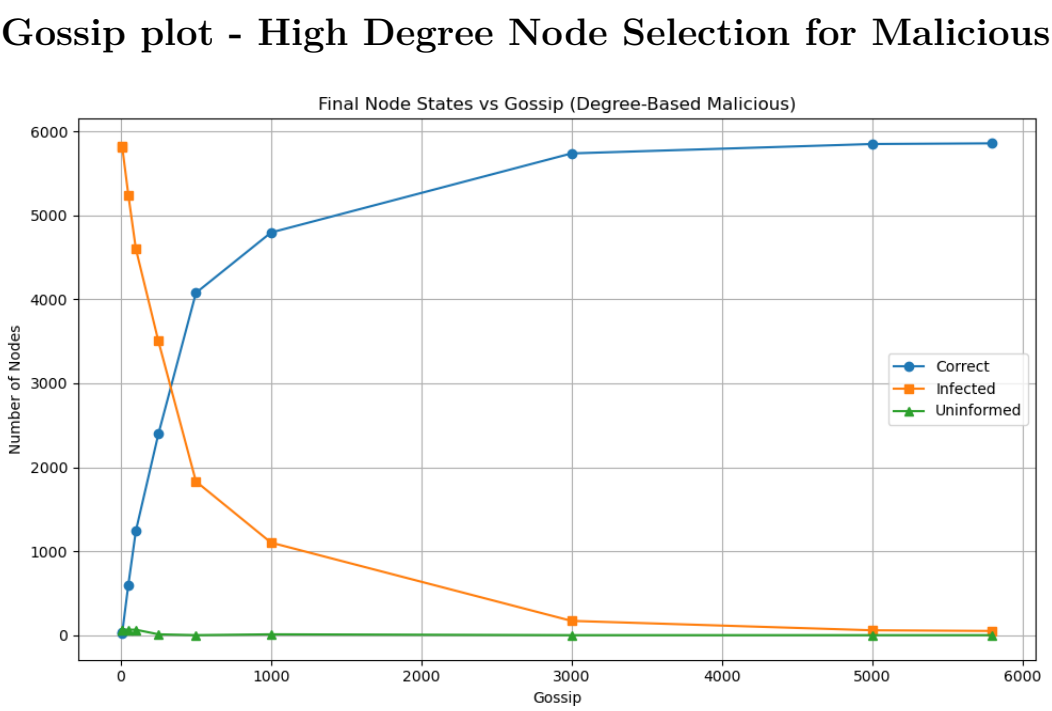


Figure 4: Information propagation with fixed θ and malicious starters, varying the number of gossip starters.

As we can observe, selecting specific nodes such as those with the **highest degree** leads to a **strong influence** within the network. This is due to the fact that such nodes can **compromise a greater number of other nodes**. Consequently, the network's resilience is **significantly compromised**, requiring a substantial increase in gossip messages to maintain security. Specifically, the network now requires between **250 and 500 gossip messages** to achieve the same level of protection that was previously obtained with only **50 messages, while the number of initial malicious nodes remains constant at 50.**

Default simulation values:

- θ: **0.15**
- Gossip starters: **30**
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Malicious plot - High Degree Node Selection for Malicious

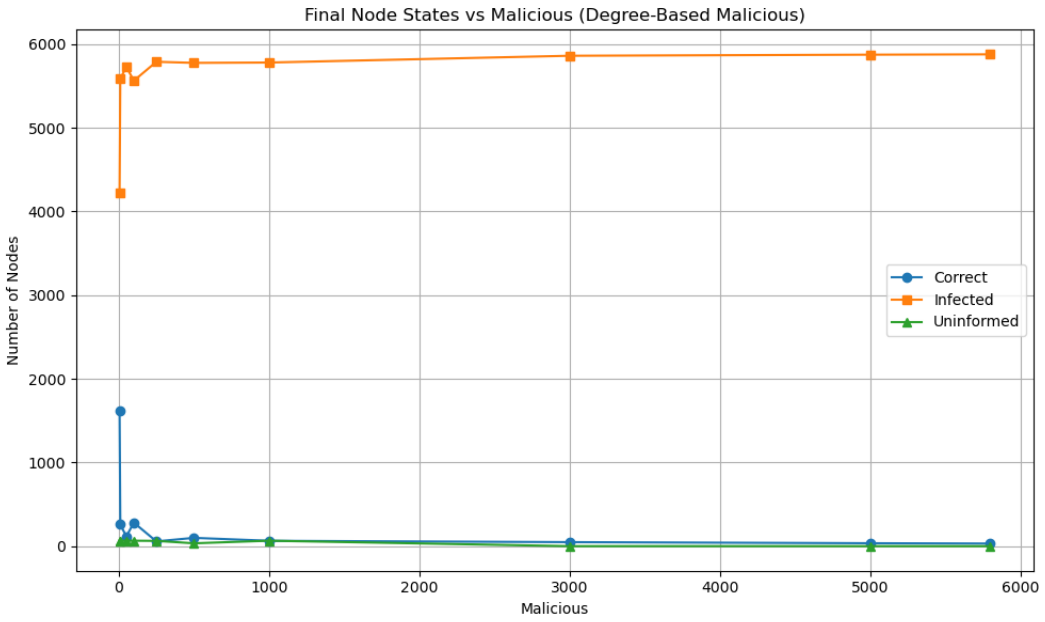


Figure 5: Information propagation with fixed θ and gossip starters, varying the number of malicious starters.

To conclude we can observe, in the case of the graph with malicious nodes, since we choose **targeted nodes**, it is sufficient to have just **5 malicious nodes** to infect **more than half of the network**, even though the gossip nodes present in the system are **6 times more numerous than the malicious ones**.

3 Considerations

After analyzing the results, I arrived at an important reflection: **nodes with high influence play a key role in the diffusion process**. The more connected a node is, the more likely it is to successfully propagate its message throughout the network.
This characteristic can be particularly exploited by **malicious nodes**, as they can **compromise the network more quickly and efficiently** if they occupy central or highly connected positions. In such roles, they participate in a larger number of local message decisions, thus **increasing their impact on the overall spread**. **If malicious nodes are not selected randomly but strategically placed on these influential positions, the diffusion of corrupted messages could accelerate significantly**, potentially requiring even fewer malicious nodes to dominate the network.
Additionally, **the structure of the network significantly affects the diffusion**. As observed, setting θ to a value that is too high can completely prevent propagation. On the other hand, a strong imbalance between the number of malicious and gossip nodes can heavily favor one side over the other.
To conclude, my reflection is that in this network, **a majority can be achieved by either side by increasing their number by just 5 or more nodes compared to the opposing side or selecting determinated starting nodes**.

4 Optional part

In the optional part, I decided to introduce a **randomized behavioral component** in the decision-making process of the nodes. Specifically, each node generates a random value between 0 and 1. If this value exceeds a predefined threshold, the node follows the majority opinion in its neighborhood; otherwise, it deliberately chooses the opposite decision.
I believe this type of simulation is particularly **interesting for modeling less rational or unpredictable behavior**, providing insights into how collective dynamics might evolve in scenarios where decisions are not purely logical or deterministic. It allows us to reflect on how real-world social systems may behave when influenced by randomness, uncertainty, or individual contrarian attitudes.

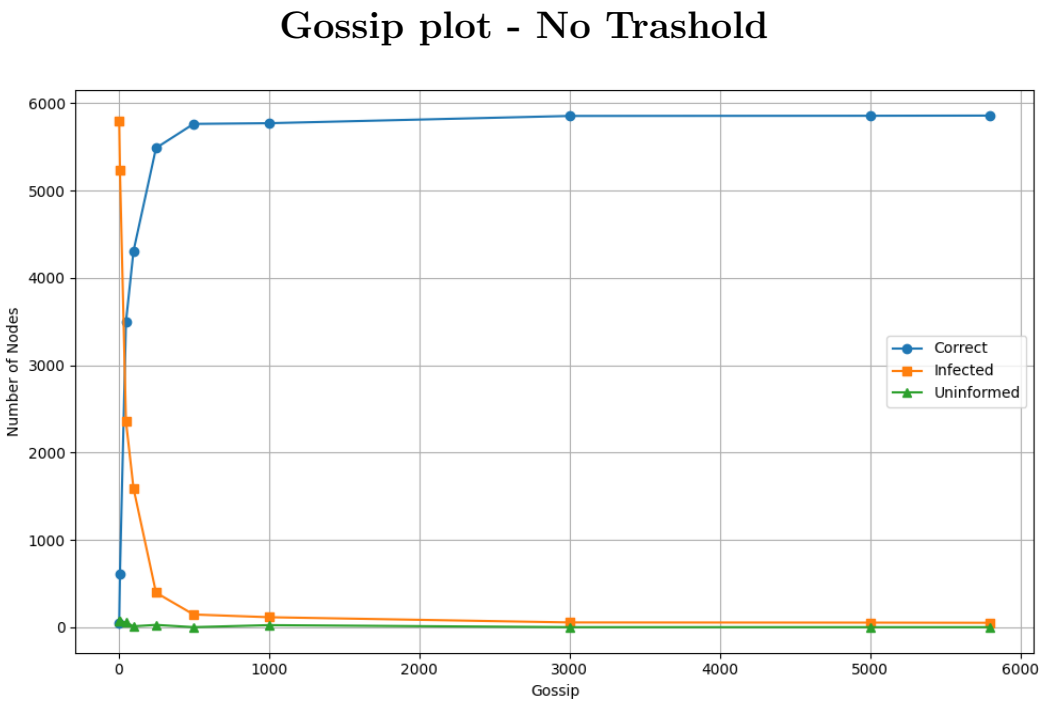


Figure 6: Information propagation with fixed θ and malicious starters, varying the number of gossip starters.

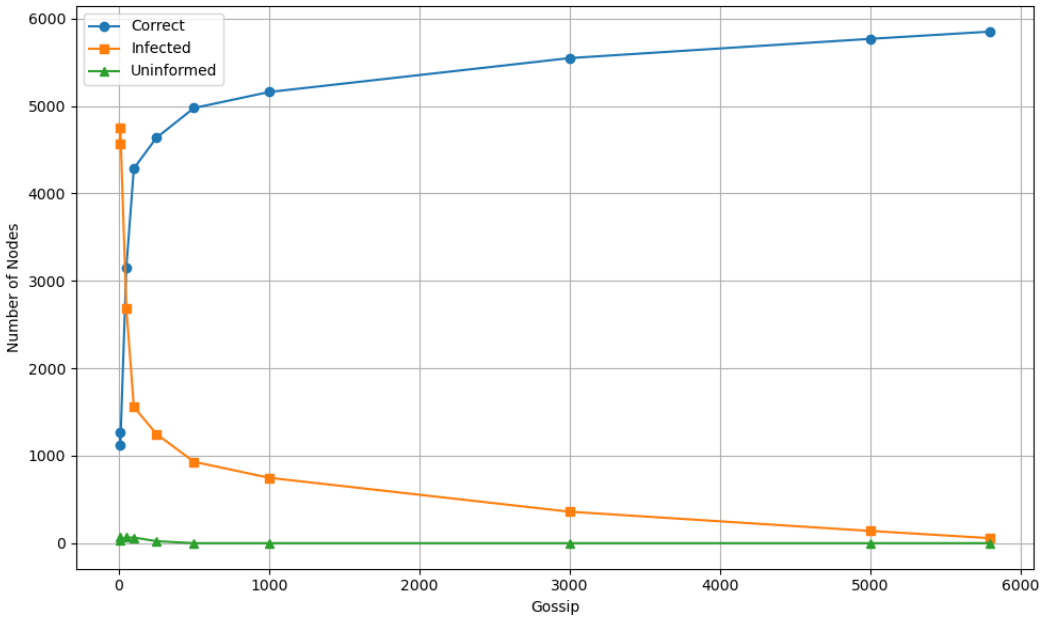


Figure 7: Information propagation with fixed θ and malicious starters, varying the number of gossip starters.

As we can observe from the two graphs, a small and abrupt change does not significantly affect the outcome of message propagation, as long as the number of starting nodes remains limited. In fact, by comparing both graphs, we notice that they each reach **a majority of the network** when starting with **50 initial nodes**. This is due to the presence of **a balance between the two parties**, namely **50 malicious nodes** and **50 gossipers**.
The real issue arises when the goal is to reach full coverage in favor of the correct message. To achieve **convergence toward correct messages**, it becomes necessary to have **almost all nodes as gossipers**, unlike in the case **without a threshold**, where **only half** would have been sufficient.
This discrepancy is mainly due to the fact that, by the time the simulation begins, **almost all nodes have already been assigned as gossipers**, leaving **very few nodes available for further propagation**. Therefore, when a node needs to make a decision, it is almost guaranteed that it will adopt the gossiper message. However, with the threshold mechanism, **every 10% of the nodes still becomes malicious** because they take an opposite decision from the rest of the group. This shift makes convergence **nearly impossible**, as the influence of malicious messages persistently disrupts the process.

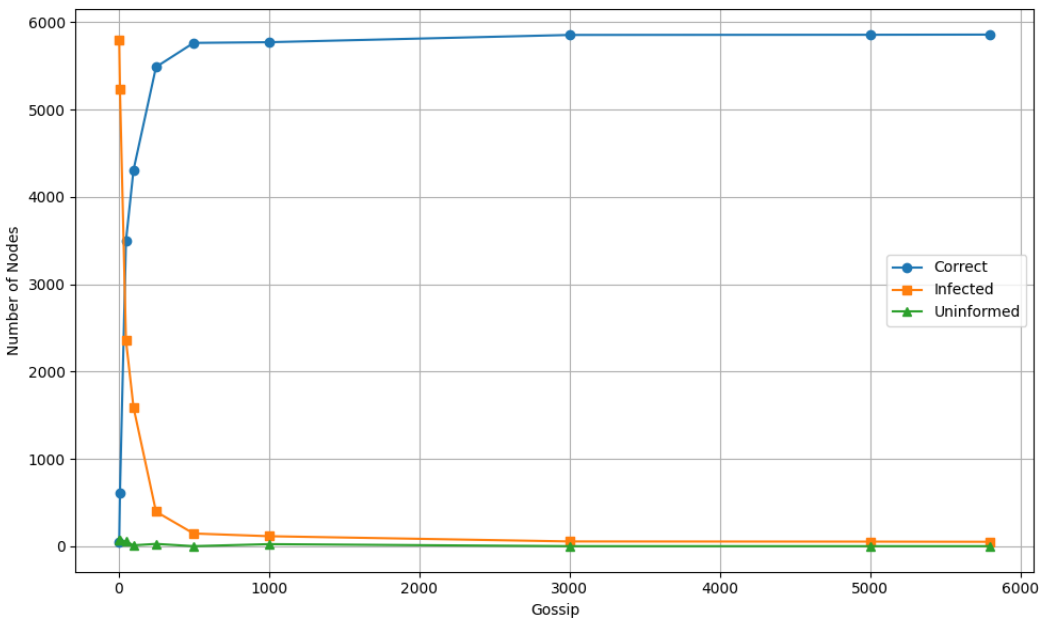


Figure 8: Information propagation with fixed θ and malicious starters, varying the number of gossip starters.

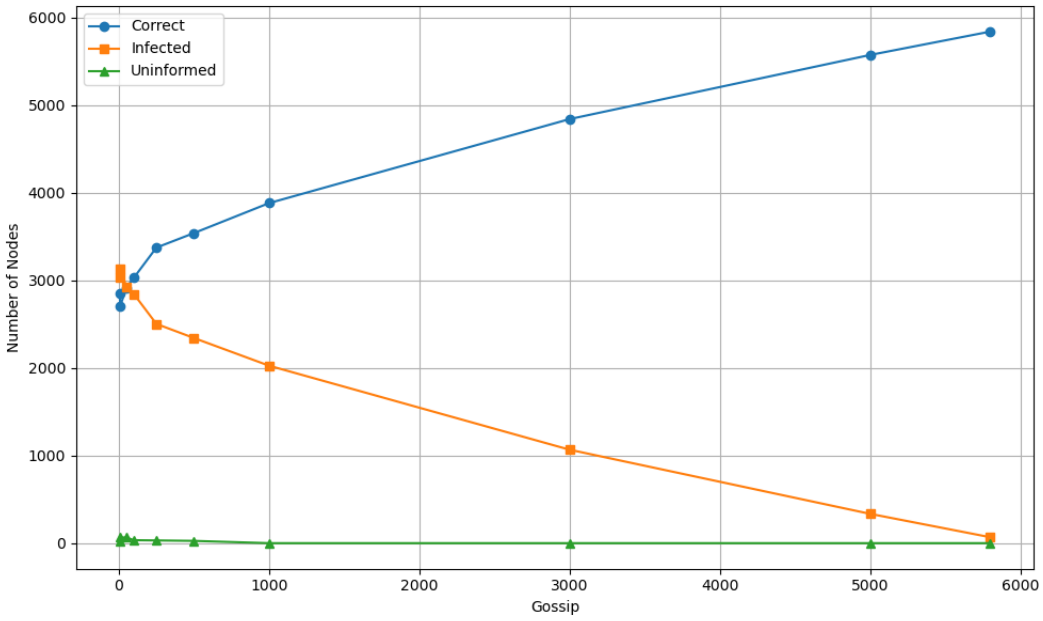


Figure 9: Information propagation with fixed θ and malicious starters, varying the number of gossip starters.

As the threshold increases, we observe that a **majority of the network** can still be reached with only **100 gossiper nodes**. However, the number of gossipers required to achieve **full convergence of correct messages throughout the entire network** rises steadily as the acceptance threshold becomes more stringent, assuming the same initial conditions as before.

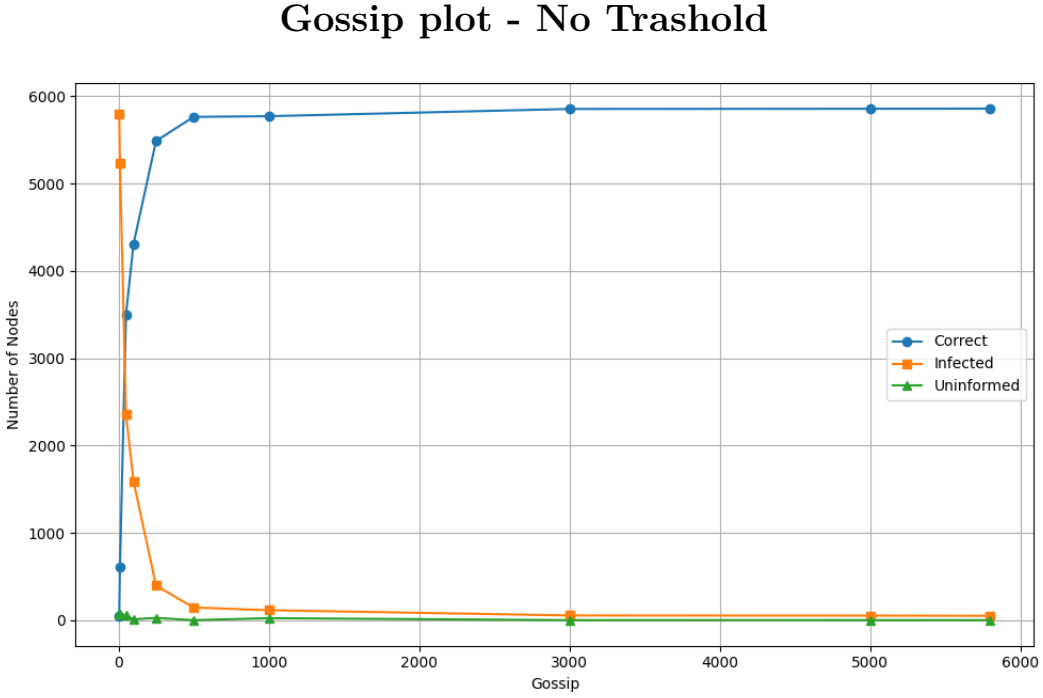


Figure 10: Information propagation with fixed θ and malicious starters, varying the number of gossip starters.

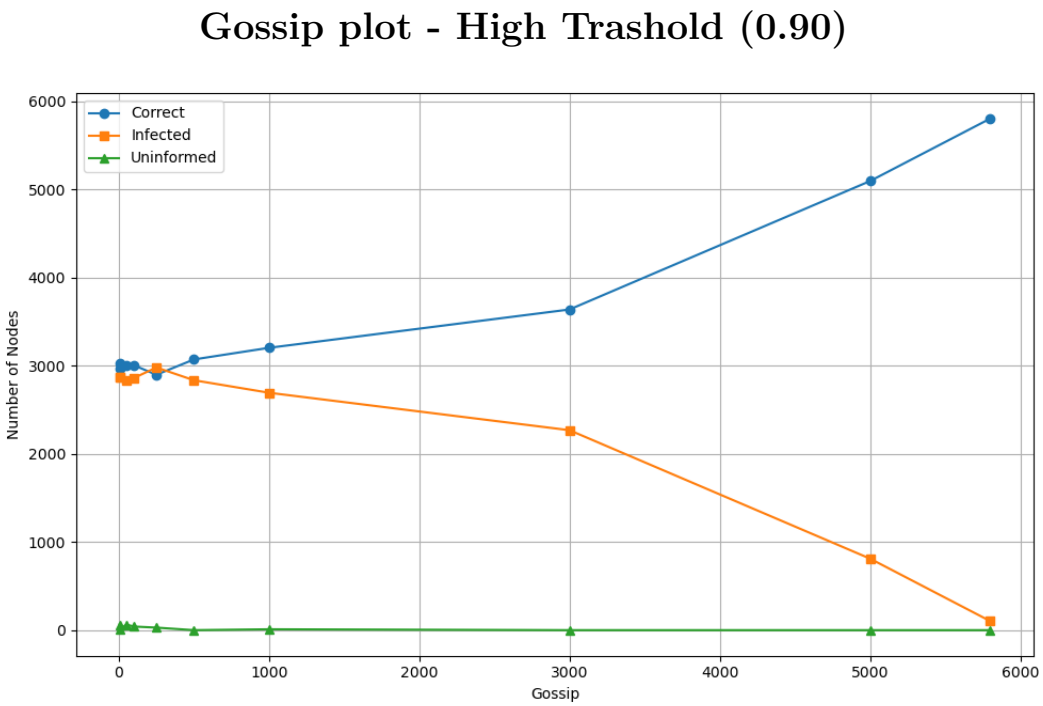


Figure 11: Information propagation with fixed θ and malicious starters, varying the number of gossip starters.

To conclude with this final graph, we can observe, as in the previous cases, that there is a significant increase in the number of initial nodes required to reach the **majority**, rising to **500 gossiper nodes**. Additionally, there is a **sharp narrowing of the margin** to achieve **full convergence**, highlighting the system's growing sensitivity to initial conditions under stricter thresholds.

5 Conclusions

Considering the previous plots, we observe that the **minimum threshold required to achieve a majority within the network** increases progressively with the **level of instability**. For instance, we identify typical values such as:

- **Low instability: 50 nodes**
- **Medium instability: 100 nodes**
- **High instability: 500 nodes**

As the threshold increases, we notice that **convergence slows down significantly**. This happens because, with a larger number of nodes of the same type, neighboring nodes tend to generate **homogeneous signals**. Under conditions of **high instability**, however, these uniform signals can become counterproductive, often leading the system to favor the **opposite direction**.
This behavior creates a natural tendency within the network to **re-balance itself**, making use of the remaining available nodes. This rebalancing effect is clearly visible in the final plot, which displays a noticeable **symmetry** in node behavior.
In summary, when many nodes act in alignment within an unstable environment, the system tends to correct itself through **emergent balancing**, resulting in a form of **dynamic equilibrium** as new nodes are introduced.